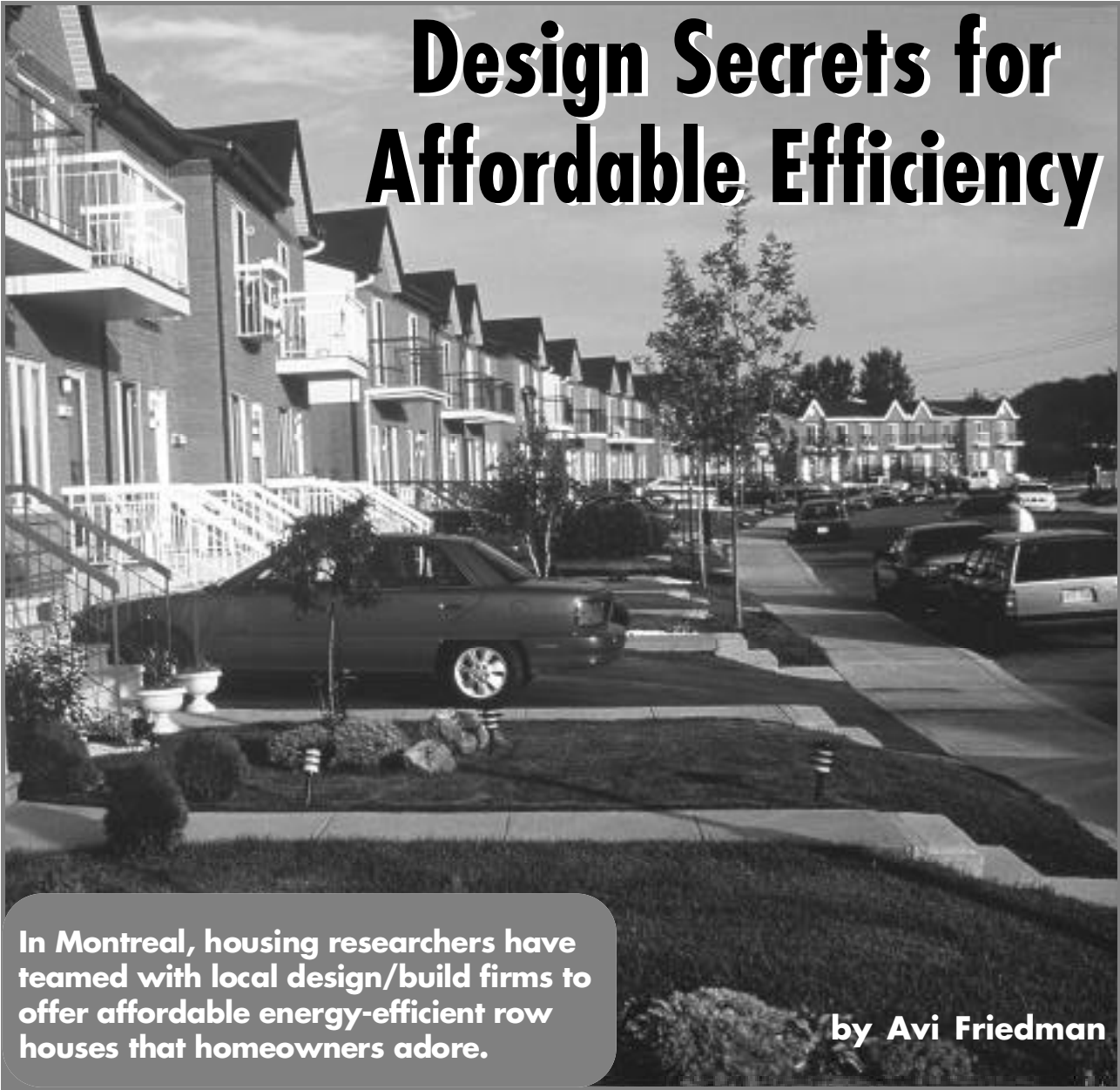




Design Secrets for Affordable Efficiency

In Montreal, housing researchers have teamed with local design/build firms to offer affordable energy-efficient row houses that homeowners adore.

by Avi Friedman

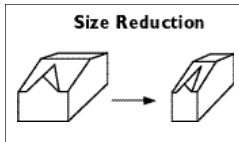
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Simply through efficient planning and appropriate construction assemblies, heat losses in homes can be reduced by half compared with typically constructed detached, single-family homes. This was my conclusion after many years of research in architecture and building technology at McGill University and other institutions. In the last ten years, these ideas have been put to the test as thousands of urban homes built to these simple design principles have been constructed and occupied in the Montreal

area. The Grow Home, a two-story, 1,000 ft², 14-ft-wide row house, was designed by myself and Witold Rybczynski (author and professor of architecture at the University of Pennsylvania) with the intention of providing an affordable urban dwelling. To date, approximately 6,000 units have been built in some 25 projects in the Montreal area (see "Growing to the Next Step," p. 27). The Grow Home experience has demonstrated that significant savings can be achieved by adhering to the following basic rules of design and construction.

Reduce the Size of the Unit

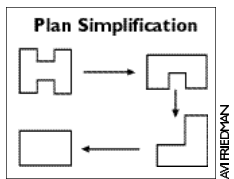
The trend in home building today is toward more rooms and larger spaces. Although some homeowners may want more space, we have found that the key to offering affordable urban dwellings is making the *best* use of the space. With efficient arrangement of interior space, a comfortable, adaptable living accommodation can be provided within just 1,000 ft² of floor area. Twelve percent of the energy used for heating and cooling the average starter home of 1,300 ft² can thus be saved through reducing



the amount of unused space that must be heated and cooled—without sacrificing occupant comfort. [Also see “Small Saves Energy (and Is Beautiful)”, *HE* July/Aug '99, p. 47.]

Simplify the Building Floor Plan

A rectangular plan for a 1,200 ft² home, for example, has about 7% less perimeter than an L-shaped plan for a home of the same area; 17% less than a U-shaped plan; and 24% less than an H-shaped plan. This reduction in building



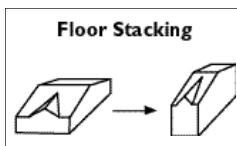
perimeter results in energy savings both directly and indirectly by reducing the exterior wall area; reducing

infiltration at the joint between the foundation wall and the exterior wall; requiring fewer or smaller windows; and requiring fewer corners, which means that less wall area is exposed to the outside, and so less infiltration occurs. In comparing estimated heat losses between two dwellings having the same floor area and fenestration, I found that approximately 15% could be saved just by simplifying the floor plan from an L shape to a rectangle. The design also encompasses a basic passive ventilation approach in that opening the windows at the front and back of the home (where all the windows are located in row houses) creates a hefty cross-breeze.

Stack the Floors on Two Levels

When the floors are stacked on two levels, although the exterior wall area is increased, the additional heat losses are usually compensated through a reduction in the roof and basement areas. Infiltration losses at the joint between the foundation wall and the exterior wall are also lowered, since the building perimeter is reduced, and fewer windows are required for the smaller basement.

The greatest advantage to stacking floors, however, comes when

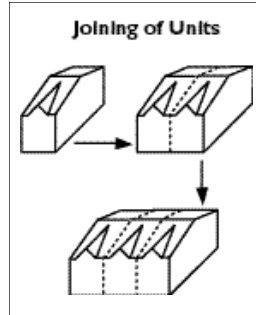


dwellings are joined in the form of duplexes or row houses. This provides significant energy savings through a major

reduction in exposed wall area, particularly when the front of the house is narrow. In the case of an 18 ft x 28 ft plan, for example, the exterior wall area is reduced by 30% for a duplex and by 60% (for all units but the end ones) for a row house. If the dimensions of the unit were changed to 14 ft x 36 ft to reduce the frontage of the house, the exposed wall areas would be 36% less for the duplex and 72% less for the row house. These reductions translate into energy savings of about 21% and 43%, respectively.

Avoid Overdesigned Structural Frames

The use of 24-inch stud spacing increases the overall thermal resistance of the wall by 4%–8%, because there is less thermal bridging, and reduces lumber costs by one-third. The choice of proper framing details at corners and intersections can also reduce the amount of wood in the wall and increase the amount of insulation. By reducing the thermal bridging at these points, the risk of condensation and eventual deterioration is also lowered.

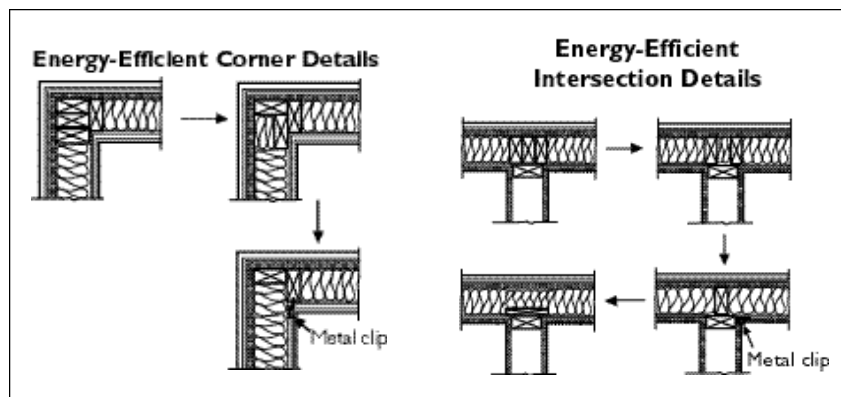


Replace the Exterior Sheathing with Rigid Foam Insulation

Replacing the exterior sheathing with 1½ inch rigid foam insulation can reduce the risk of thermal bridging and increase the thermal resistance of the wall by 33%, although this is not recommended where the sheathing is needed for structural strength. It can have a significant effect in cold climates. Where the thickness of the wall is critical, as is the case in narrow-front row housing (where every inch counts) this sheathing can be fastened to a 2 x 4 stud frame. The overall thermal resistance of the wall would be the same as that of the 2 x 6 frame, but the wall would be slightly narrower and the assembly process would be lighter and possibly quicker.

Ensure Good Workmanship

Poor workmanship can substantially affect the thermal performance of the wall section. If 3½ inch batt insulation is compressed by ¼ inch through improper installation techniques, for example, its thermal resistance is reduced by about 4%. Air infiltration at the joints between the window or door frames and the exterior wall, combined with those at the joint between the foundation wall and the exterior wall, account for about 20% of heat losses in houses with average workmanship. Improperly installed flashing and non-continuous insulation in these areas not only will seriously affect the thermal performance of the wall but also will increase the risk of condensation inside the wall, which will eventually deteriorate the insulation and reduce the wall's



Energy-efficient corner details and intersection details should always be used for greatest savings.

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performance even further. The problem is made worse when the vapor barrier on the warm side is hastily installed or is not continuous. All joints should be overlapped and caulked to ensure the barrier's integrity. Frequent inspections—sometimes unannounced—by the designer, city inspectors, and the builders ensure good workmanship in the Grow Homes.

Minimize the Number of Openings in the Vapor Barrier

To minimize the number of openings in the vapor barrier that require cutting and overlapping of the membrane, (1) minimize the number of electrical boxes on the exterior wall, (2) avoid placing plumbing in the exterior wall, and (3) avoid recessed lighting fixtures that require deep electrical boxes in the upper-floor ceiling.

Carefully Select and Properly Orient Windows

One of the most effective ways of controlling heat loss is through the careful selection and proper orientation of windows. In the case of 14-ft row housing,

heat losses from the wall and roof section account for about 30% of total heat loss, while losses from windows and infiltration account for 44% of the total. In North America, considerable heat gains can be achieved by orienting more windows toward the south. Windows should be installed carefully, and rough openings should be thoroughly sealed.

Three additional aspects of windows should also be considered. These are the type of window, the type of frame, and the type of glazing.

Type of window. Windows that require a sliding assembly are generally less efficient than windows that require a pivotal mechanism, and fewer working parts mean fewer linear feet of seam through which infiltration can occur. Infiltration is responsible for approximately 25% of all energy loss related to windows. Fixed, awning, and casement windows therefore tend to perform better than either double-hung or horizontal sliders. The effective leakage area of casement windows, for example, is about one-third less than that of a horizontal slider and one-quarter less than that of a double-hung unit of the same size. This is

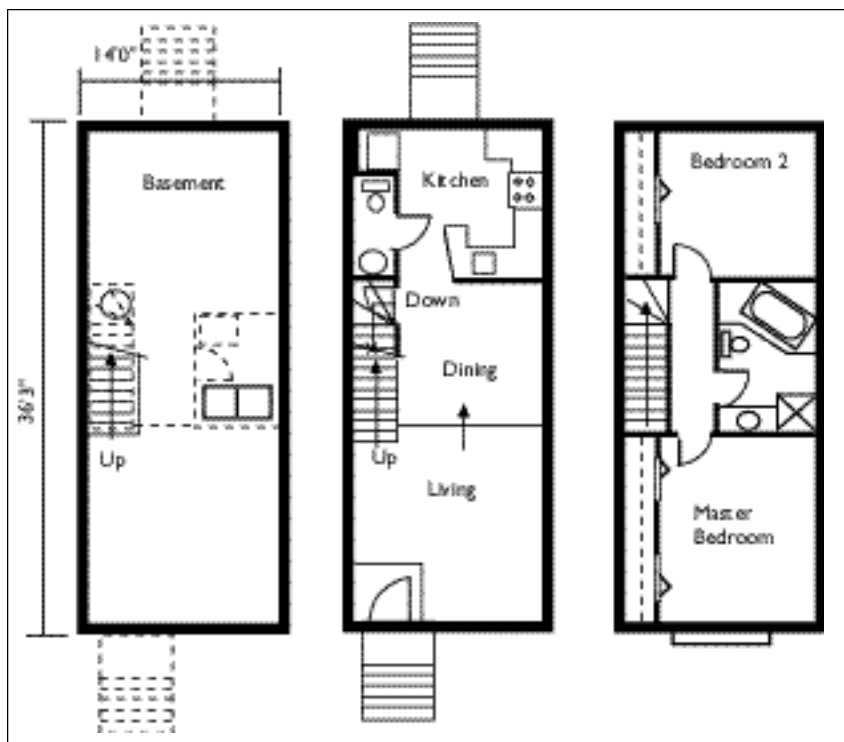
true because many of the horizontal slider windows are cheaply made, and with the double-hung windows, there are more linear feet of window edge to allow air infiltration.

Type of frame. The material of the window frame also has an effect on the overall thermal performance of the unit. Wood frames are usually more effective insulators, followed by vinyl, thermally broken metal, and metal frames. Wood or vinyl frames can reduce the thermal conductance of the window unit by 23% over units with thermally broken aluminum frames. Metal frames with no thermal break should not be used in energy-efficient building designs. All units should be weatherstripped at the factory.

Type of glazing. The upgrade from single-glazed units to double glazing promises the highest savings; it improves thermal performance by 40%–55%. The addition of low-e coatings increases thermal performance by an additional 15%–25%. Because so much heat loss in houses occurs through the windows, these figures can provide substantial overall energy savings. In a 1,000 ft² dwelling, for instance, the upgrade from single- to double-glazed units can represent savings of about 17%. The addition of a low-e coating can further reduce heating costs by 9% for a detached house, 11% for a duplex, and 16% for a row house. The percentage of savings increases with the reduction in window area. Upgrading the units with argon-filled glazing provides only about 2% additional savings. Since the average cost of these windows in Canada is about CAN\$150, the cost-effectiveness of this last upgrade is questionable.

Consider a Basement

The decision whether or not to include a basement in the house will be influenced to some degree by the climate in which it is built. In cold climates, building codes may require that the foundation walls be constructed to a depth of 4 ft or 4½ ft below grade to ensure that the footings rest below the frost line. If this is the case, a basement could be provided at a small marginal cost to increase the floor area by one-third. For narrow-front row houses, heat



A small house can be a comfortable home, as this Grow Home floor plan shows.

Growing to the Next Step

During the period between 1990 to the end of 1999, more than 6,000 Grow Home units were constructed on the same basic model, although all of the builders modified the design slightly to suit the tastes and budgets of their own particular markets.

The houses ranged in price from CAN\$69,000 to CAN\$95,000. Other average homes in the Montreal area range

home that can easily be modified to function as a fully detached home, as a duplex, or as a row house. The four levels of the Next Home structure can not only be arranged in a manner that suits the present requirements of its residents but can also be re-arranged at a future date to accommodate household and family changes. Key features of the Next Home project include: buying only the quantity of space that the user needs and can afford, making the houses affordable, allowing the residents to design the interior layout by selecting from a catalog of components, maintaining the flexibility to change the interior layout choice of facade design, ensuring both environmental responsibility and comfort, considering export potential, and adopting a new urban perspective.

As part of the design of the Next Home facade, the user in a multi-unit

structure can choose from a range of fenestration and door accessories, provided by the builder and designed by an architect, that will determine the final visual appearance of the openings. Once the openings have been selected, the user chooses from a variety of window and door options to complete the design of the opening and to create a personal facade for the unit. Such an element of choice not only provides personalization but also creates visual diversification in a row of Next Home structures.

The occupants will also be able to choose from a catalogue of interior components to suit their individual lifestyles and budgets. These components (kitchen, bathroom, and so on) are selected at the pre-construction stage and allow future residents to "consume" only those elements they require.

Affordability continues to be a major impediment to homeownership for many people. The Next Home—which offers prospective buyers close to 700 square feet of living space for only CAN\$50,000 (including land) in a city such as Montreal—is a long-awaited solution to this crisis in the housing market.



The unique, modular design of the Next Home allows it to be built as a detached home, a duplex, or a triplex. The triplex shown above is located in Longueuil, Quebec.



from about CAN \$110,000–120,000. No Grow Homes have been built so far in the United States, but some pre-fabricated houses based on the design will soon be available in the Northeast.

All the Grow Homes sold quickly, remain occupied, and have retained their value in spite of the depressed economy. Although they were originally marketed as starter homes, the builders found that the average income of the Grow Home buyers was higher than expected—many people who could afford to do so simply didn't want to spend a larger portion of their incomes on housing. Many of the new owners modified the unfinished space in the basement to suit their needs. Surveys showed that occupants were very satisfied with the homes. Furthermore, very few of the homes have been resold—the occupants liked them too much to leave!

The Grow Home was recognized with a World Habitat Award by the United Nations on October 4, 1999 (World Habitat Day), at a ceremony in Dalian, China, for its affordability and replicability features.

A second prototype, the Next Home, extends the Grow Home design into a

losses would be increased by about 11%. Because the heating costs for this type of housing are relatively low, this increase will not add much to annual expenses. Considering that a basement will increase the total living area by approximately one-third, the small added expense may prove to be worthwhile.

Conclusion

My experience with the design of affordable housing has shown that energy-efficient dwellings can be built at a reasonable price. Simple and efficient space planning eliminates the cost of heating unused space and reduces heat loss through exposed wall sections. Designing for ease of assembly with conventional building materials makes the quality of workmanship easier to control and reduces the risk of infiltration from inadequate assembly of complex building details. The use of high-quality windows provides a simple and effective way of substantially reducing heat losses, from both conduction and infiltration.

Above all, my experience shows that affordable housing can be built without sacrificing either quality or energy efficiency. Common sense planning, simple detailing, and careful material selection are as effective as any other means when it comes to cutting building and heating costs.

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